

MIDTERM PROJECT GUIDELINE



1. **SCOPE** - This mid-term project requires students to address and fulfill the pipeline of Data Analysis. Through which process, the project aims to answer given questions by a given dataset.
2. **DELIVERY** - The project is marked as successfully submitted when students completely deliver 3 components:
 - a. A Google Colaboratory Notebook that contains a Python script to load, clean, (transform, manipulate), analyze, and visualize data.
 - b. A group presentation in 15 minutes, which is participated by all the members of the team.

3. TIMELINE

- Presentation Day: Friday 09/08/2024
- Deadline to submit all the related documents: 23:59 Sunday 11/08/2024

4. DATASETS

In this project, students are given 5 datasets.

To narrow down the scope of the project while maintaining the diversity of final presentations, each group gets to choose one topic/dataset.

1. **Global Terrorism Data:** <https://www.kaggle.com/START-UMD/gtd>
2. **Learn Platform Covid19 Impact on Digital Learning:**
<https://www.kaggle.com/competitions/learnplatform-covid19-impact-on-digital-learning/data>
3. **2021 Kaggle Machine Learning and Data Science Survey:**
<https://www.kaggle.com/c/kaggle-survey-2021>
4. **Brazilian Public Dataset by Olist:**
<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>
5. **Airbnb Dataset:**
<https://insideairbnb.com/get-the-data/>

5. **EVALUATION:** The evaluation of the project only starts once students complete the submission of their final project by the deadline, which requires 3 components (as written in section 2). Failure to deliver any of the 3 components will result in **NOT SUBMIT** and affect the graduation status.

On a scale of 15, the project is evaluated as the following criteria:

FEATURE	BAND	DESCRIPTION
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IDEA <i>(max 3)</i>	3	Able to deliver a coherent story that answers the pre-defined topic. The story is supported with strong evidence from data and external research.
	2	Able to deliver a story that answers the questions. Yet, the insights are not pursued comprehensively.
	1	Able to find insights but fail to put them as a story.
TECHNICALITY <i>(max 3)</i>	3	Output from code helps answer the questions. Clean code.
	2	There are still minor insights that are not justified by the output of the code.
	1	The output of the code is irrelevant with the insight. Forcing the output to support a pre-defined insight when it is naturally not.
Q&A <i>(max 3)</i>	3	Able to deliver accurate answers in their own language with supporting evidence from the analysis.
	2	Minor mistakes but able to deliver the correct explanation.
	1	Students may make mistakes in questions that are not fundamental (for example meaning of syntax, simple technical terms).
DELIVERY <i>(max 3)</i>	3	Able to deliver the story on time. The presentation has a good structure and is easy to follow. All the members participate in the presentation.
	2	Able to deliver the story on time. There are still some obscure points that don't link with the storyline.
	1	Poorly structured presentation.
BONUS <i>(max 3)</i>	*	The 3 bonus points are for but not limited to: - Using creative solutions to deliver the analysis (new forms of presentation, supporting material, etc) - Significant work on feature engineering, data transformation, or incorporating external data sources. - Incorporate machine learning to extend the analysis.

* A 0 (zero) means a failure to deliver the evaluation feature.